

6.622 Power Electronics Design Project

Synchronous Buck Converter

Matthew De Jesus

May 12, 2026

Design Summary

Table 1: Selected components and key operating parameters.

Ref	Element	Part / configuration	Key specs
<i>Power stage</i>			
Q_1	High-side MOSFET	onsemi NVMFS5844NL	60 V, 12 m Ω , 35 ns t_r+t_f
Q_2	Low-side MOSFET	onsemi NVMFS5C628NL	60 V, 2.4 m Ω , 150 A
L_1	Output inductor	RM12/I-3F3-A400 + bobbin + 6T 13 AWG	14.4 μ H, $B_{pk} = 0.267$ T
C_1	Output capacitor	10 $\times$ 106K100CS4 (parallel)	100 μ F, 0.3 m Ω ESR _{eff}
<i>Input filter</i>			
C_2	Input cap bank	14 $\times$ KTS500B335M32N0T00 (parallel)	46 μ F, 0.71 m Ω ESR _{eff}
L_2	Input choke	RM10/I-3F3 + bobbin + 2T 16 AWG	16.2 μ H
<i>Operating point</i>			
f_{sw}	Switching frequency	100 kHz, synchronous	
ΔI_L	Output inductor ripple	5.83 A pp at $V_{in} = 40$ V	
BOM cost	Estimated cost	\$513.27	
ΔT	Heat-sink	None (Not Required)	

Table 2: Predicted performance vs. specification requirements (worst-case corner).

#	Specification	Required	Predicted
1	Output voltage (static)	12 V $\pm$ 3 %	12 V
2	Output voltage ripple	$\leq$ 75 mV pp	74.7 mV pp
3	Output transient ($\pm$ 120 W step)	$\pm$ 20 % ($\pm$ 2.4 V)	2V
4	Input current ripple	$\leq$ 60 mA pp	56.6 mA pp
5	Minimum efficiency	$\geq$ 90 %	97.81 % at full load & $V_{in,max}$
6	MOSFET junction temperature	$\leq$ 150 $^{\circ}$ Celsius	139.38 $^{\circ}$ Celsius
7	Inductor saturation	$B_{pk} \leq$ 0.30 T	0.267 T (1.12 $\times$ margin)

1 Motivation And Python Optimization

To mimic a real-life design challenge, I imagined designing this buck converter for MIT’s Solar Car team. They have a limited amount of solar panel space, so they need to push for the highest efficiency, and cost is not a concern. To find the most efficient buck converter parameters, I used a Python script to optimize for efficiency using the design constraints described throughout the component selection sections and the loss equations found in Section 7 [1]. However, what I found was that an efficient enough power converter has very little power loss through its MOSFETs, enough for the converter to do away with a heat sink. The no-heat-sink MOSFET loss constraint is

$$P_{device,max} = \frac{T_{J,max} - T_{amb,max}}{\theta_{JA}} = \frac{150^\circ\text{C} - 50^\circ\text{C}}{40^\circ\text{C/W}} = 2.5\text{ W}. \quad (1)$$

Therefore, each individual MOSFET must dissipate less than approximately 2.5 W at the worst operating point.

2 Topology and Operating Point

The converter is a synchronous buck with a single-stage LC input filter ahead of the converter input capacitance. A synchronous buck was chosen over a Schottky diode for the low-side device to maximize efficiency: the low-side current path conducts for $1-D$ of every cycle (70% at $V_{in} = 40\text{ V}$) at an RMS current of 11.16 A, and the 0.5 V forward drop of a Schottky in this position dissipates 4.67 W versus the 0.45 W of conduction loss in the 2.4 mΩ synchronous rectifier.

The worst-case operating corners that drive component selection are:

- Switching-loss worst case: $V_{in} = 40\text{ V}$, $P_{out} = 160\text{ W}$. $D = 0.30$; HS sees full bus voltage across switching transitions.
- Conduction-loss worst case: $V_{in} = 27\text{ V}$, $P_{out} = 160\text{ W}$. $D = 0.444$; maximizes HS RMS current and input-cap RMS current.
- Input current ripple worst case: $V_{in} = 27\text{ V}$ (duty closest to 0.5, where $I_{rms,in} = I_{out}\sqrt{D(1-D)}$ peaks).

The switching frequency was selected at $f_{sw} = 100\text{ kHz}$ to minimize switching losses while staying in continuous-conduction mode (CCM) at the 40 W minimum load. The CCM/DCM boundary requires $\Delta I_L \leq 2 I_{out,min}$, i.e. $\Delta I_L \leq 6.67\text{ A}$; the chosen design has $\Delta I_L = 5.83\text{ A}$.

3 Power Device Selection

3.1 MOSFET selection: HS–LS asymmetry

The two MOSFET positions are operated very differently and have opposite optimization targets:

- The **HS is hard-switched**. During turn-on, V_{DS} falls from V_{in} to zero while I_D rises from 0 to I_{load} ; the $V \cdot I$ overlap dissipates $\frac{1}{2} V_{in} I_{out} (t_r + t_f) f_{sw}$ per cycle. Switching loss therefore favors low Q_g (Gate Charge) and fast transitions.
- The **LS is soft-switched (ZVS)**. The body diode pre-conducts during the dead time, so $V_{DS} \approx 0$ when the channel turns on. Switching loss is negligible; only conduction loss matters, favoring lowest $R_{DS(on)}$.

Table 3: Candidate MOSFETs (max $R_{\text{DS(on)}}$ at $V_{\text{GS}} = 10 \text{ V}$).

Part	$R_{\text{DS(on)}}$ (m Ω)	Q_g (nC)	$t_r + t_f$ (ns)	C_{oss} (pF)	V_F (V)
NVMFS5826NL	24	17	56	85	0.80
NVMFS5844NL	12	30	35	150	0.79
NVMFS5C670NL	6.1	20	64	690	0.90
NVMFS5C628NL	2.4	52	63.5	1700	0.80

Table 3 summarizes the four candidate MOSFETs provided in the project component set.

To find the optimal pair, all $4 \times 4 = 16$ (HS, LS) combinations were evaluated using the loss model in Section 7. The full sweep gives:

- The LS choice dominates total efficiency — swapping the LS device changes η by up to 3.0 pp; swapping the HS device by at most 1.0 pp.
- The HS ranking is identical at every LS choice: $5844 > 5C628 > 5C670 > 5826$. The 5844NL wins because its 25/10 ns rise/fall combined with 12 m Ω $R_{\text{DS(on)}}$ is the best balance.
- The LS ranking is also consistent: $5C628 > 5C670 > 5844 > 5826$. Lower $R_{\text{DS(on)}}$ always wins on the soft-switched side.

The optimal pair is NVMFS5844NL on the HS and NVMFS5C628NL on the LS, with predicted total MOSFET loss 2.74 W at full load — a 1.62 W improvement over the same HS–LS FET design and a 4.93 W improvement over the non-synchronous Schottky alternative.

3.2 Loss model

For the HS device, switching loss is computed assuming linear current rise and fall and is summed with conduction, gate-drive, output-capacitance, and reverse-recovery loss: [1].

$$P_{\text{cond,HS}} = I_{\text{rms,HS}}^2 \cdot R_{\text{DS(on)}} \cdot k_T = (7.30 \text{ A})^2 \cdot 12 \text{ m}\Omega \cdot 1.5 = 0.96 \text{ W} \quad (2)$$

$$P_{\text{sw,HS}} = \frac{1}{2} V_{\text{in}} I_{\text{out}} (t_r + t_f) f_{\text{sw}} = \frac{1}{2} \cdot 40 \cdot 13.33 \cdot 35 \text{ ns} \cdot 100 \text{ kHz} = 0.933 \text{ W} \quad (3)$$

$$P_{\text{drv,HS}} = Q_g V_{\text{GS}} f_{\text{sw}} = 30 \cdot 10 \cdot 10^{-9} \cdot 10^5 = 0.030 \text{ W} \quad (4)$$

$$P_{\text{coss,HS}} = \frac{1}{2} C_{\text{oss}} V_{\text{in}}^2 f_{\text{sw}} = 0.012 \text{ W} \quad (5)$$

$$P_{Q_{\text{RR,HS}}} = Q_{\text{RR},Q_2} V_{\text{in}} f_{\text{sw}} = 60 \cdot 40 \cdot 10^{-9} \cdot 10^5 = 0.240 \text{ W} \quad (6)$$

$$\Rightarrow P_{\text{tot},Q_1} = 2.18 \text{ W} \quad (7)$$

For the LS device, $I_{\text{rms,LS}} = I_{\text{out}} \sqrt{1 - D} = 11.16 \text{ A}$, and switching loss is taken as zero (ZVS):

$$P_{\text{cond,LS}} = I_{\text{rms,LS}}^2 \cdot R_{\text{DS(on)}} \cdot k_T = (11.16 \text{ A})^2 \cdot 2.4 \text{ m}\Omega \cdot 1.5 = 0.448 \text{ W} \quad (8)$$

$$P_{\text{drv,LS}} = Q_g V_{\text{GS}} f_{\text{sw}} = 0.052 \text{ W} \quad (9)$$

$$P_{\text{dt,LS}} = V_F I_{\text{out}} t_{\text{dead}} \cdot 2 \cdot f_{\text{sw}} = 0.8 \cdot 13.33 \cdot 30 \text{ ns} \cdot 2 \cdot 10^5 = 0.064 \text{ W} \quad (10)$$

$$\Rightarrow P_{\text{tot},Q_2} = 0.564 \text{ W} \quad (11)$$

I used the on-resistance temperature factor $k_T = 1.5$ accounts for the $\sim 50\%$ rise in $R_{\text{DS(on)}}$ at the expected $T_J \approx 83^\circ \text{Celsius}$, since the device junctions operate at a higher temperature without a heatsink.

3.3 Thermal analysis and heatsinking

Both MOSFETs have a similar junction to case thermal resistance (datasheet $R_{\theta JA} = 15.1^\circ\text{Celsius/W}$). With this we can calculate the maximum temperature.

$$T_{J,Q_1} = T_{\text{amb}} + P_{\text{tot},Q_1} \cdot R_{\theta JA} = 50 + 2.18 \cdot 41 = 139.38^\circ\text{Celsius} \quad (12)$$

This is below the 150°Celsius ceiling but provides 10°Celsius of margin. However with thermal vias and a large copper pour, the PCB acts as the devices heat sink improving the thermal margin further.

4 Output Inductor Design

The output inductor was sized for a target ripple $\Delta I_L = 5.83\text{ A}$ at the worst-case. We get the following equations:

$$L_1 = \frac{(V_{\text{in,max}} - V_{\text{out}}) V_{\text{out}}}{V_{\text{in,max}} f_{\text{sw}} \Delta I_L} = \frac{(40 - 12) \cdot 12}{40 \cdot 10^5 \cdot 5.83} = 14.4\text{ }\mu\text{H} \quad (13)$$

Among the provided Philips RM cores in 3F3 ferrite, the smallest core whose bobbin can accommodate the required copper is RM12/I (winding area $W_a = 75\text{ mm}^2$). Using the A400 gapped variant ($A_L = 400\text{ nH}$):

$$N = \left\lceil \sqrt{\frac{L_1}{A_L}} \right\rceil = \lceil \sqrt{36} \rceil = 6 \text{ turns}, \quad L_{\text{actual}} = N^2 A_L = 14.4\text{ }\mu\text{H} \quad (14)$$

The peak inductor current $I_{\text{pk}} = I_{\text{out}} + \Delta I_L/2 = 16.25\text{ A}$ gives a peak flux density:

$$B_{\text{pk}} = \frac{N A_L I_{\text{pk}}}{A_e} = \frac{6 \cdot 400\text{ nH} \cdot 16.25\text{ A}}{146\text{ mm}^2} = 0.267\text{ T} \quad (15)$$

This is well below the project's 0.30 T 3F3 saturation limit (a $1.12\times$ margin). The RMS inductor current $I_{\text{rms,L}} = \sqrt{I_{\text{out}}^2 + \Delta I_L^2/12} = 13.44\text{ A}$ requires a winding cross-section of

$$A_w = I_{\text{rms,L}}/J_{\text{max}} = 13.44/5 = 2.69\text{ mm}^2 \quad (16)$$

at the project-limited current density $J_{\text{max}} = 500\text{ A/cm}^2$. This is 13 AWG ($\varnothing 1.83\text{ mm}$) magnet wire. The window fill ratio is

$$\text{fill} = \frac{N A_w}{k_u W_a} = \frac{6 \cdot 2.69}{0.30 \cdot 75} = 0.72 \quad (17)$$

which fits within the bobbin with no margin to spare. The winding DC resistance is

$$R_{dc,L_1} = \frac{\rho_{\text{Cu}} N \ell_{\text{avg}}}{A_w} = \frac{2.3 \times 10^{-8} \cdot 6 \cdot 61\text{ mm}}{2.69\text{ mm}^2} = 3.15\text{ m}\Omega \quad (18)$$

giving winding loss $P_{Cu,L_1} = I_{\text{rms,L}}^2 R_{dc} = 0.57\text{ W}$. For core loss, the peak AC flux is

$$B_{\text{ac,pk}} = \frac{L_1 \Delta I_L}{2 N A_e} = \frac{14.4 \cdot 10^{-6} \cdot 5.83}{2 \cdot 6 \cdot 146 \cdot 10^{-6}} = 0.048\text{ T} \quad (19)$$

and the Steinmetz fit for 3F3 in the 100–300 kHz band gives:

$$P_{\text{core}} = C_M f^\alpha B_{\text{ac,pk}}^\beta V_e = 2.5 \times 10^{-4} \cdot 10^{5 \cdot 1.63} \cdot 0.048^{2.45} \cdot 8.34\text{ cm}^3 = 173\text{ mW} \quad (20)$$

Centerpost temperature rise: $\Delta T = (P_{Cu} + P_{\text{core}}) \cdot R_{\theta,\text{RM12}} = 0.74 \cdot 23 = 17^\circ\text{Celsius}$, well below the 50°Celsius guideline.

5 Output Capacitor

The output capacitor must hold output ripple to ≤ 75 mV. For an inductor-driven triangular ripple current, we use the following equation to calculate the worst-case peak-to-peak voltage ripple.

$$\Delta V_{\text{out,pp}} = \Delta I_L \cdot \text{ESR}_{\text{eff}} + \frac{\Delta I_L}{8 f_{\text{sw}} C_{\text{eff}}} \quad (21)$$

Ten 106K100CS4 capacitors in parallel give $C_{\text{eff}} = 100 \mu\text{F}$ and $\text{ESR}_{\text{eff}} = 3 \text{ m}\Omega/10 = 0.3 \text{ m}\Omega$:

$$\Delta V_{\text{out,pp}} = 5.83 \cdot 0.3 + \frac{5.83}{8 \cdot 10^5 \cdot 100 \cdot 10^{-6}} \cdot 10^3 = 1.7 + 72.9 = 74.7 \text{ mV} \quad (22)$$

This leaves 0.3 mV of margin under the 75 mV spec. The output cap RMS current is $I_{\text{rms,out}} = \Delta I_L / \sqrt{12} = 1.68 \text{ A}$, well under the combined ripple rating. Conductive-polymer material was selected for its low, flat ESR across frequency.

6 Input Filter

The input filter must attenuate the converter's pulsed input current (amplitude $\approx 16.7 \text{ A}$ pp at f_{sw}) down to the 60 mA pp source-side spec, approximately $20 \log_{10}(16.7/0.060) \approx 49 \text{ dB}$ of attenuation.

6.1 Input capacitor bank

The input cap sees the highest converter-side ripple current, equal to $I_{\text{rms,in}} = I_{\text{out}} \sqrt{D(1-D)}$. The worst case is at $V_{\text{in,min}}$ (where $D = 0.444$, closest to 0.5):

$$I_{\text{rms,in}} = 13.33 \sqrt{0.444 \cdot 0.556} = 6.62 \text{ A} \quad (23)$$

Fourteen KTS500B335M32N0T00 caps in parallel provide $46 \mu\text{F}$, an effective ESR of $0.71 \text{ m}\Omega$, and combined ripple-current capability. $P_{C_2} = I_{\text{rms,in}}^2 \cdot \text{ESR}_{\text{eff}} = 0.031 \text{ W}$, split across fourteen caps.

6.2 Input choke

With the bank providing $46 \mu\text{F}$ of capacitance, the input filter inductance required to achieve the 49 dB attenuation target is:

$$\frac{f_{\text{sw}}}{f_r} \geq 16.7 \Rightarrow f_r \leq 5.99 \text{ kHz} \Rightarrow L_2 \geq \frac{1}{(2\pi f_r)^2 C_2} = 15.3 \mu\text{H} \quad (24)$$

The RM10/I-3F3 with 2 turns gives $L_2 = N^2 A_L = 16.2 \mu\text{H}$:

$$f_r = \frac{1}{2\pi \sqrt{L_2 C_2}} = \frac{1}{2\pi \sqrt{16.2 \cdot 10^{-6} \cdot 46 \cdot 10^{-6}}} = 5.8 \text{ kHz} \quad (25)$$

giving attenuation of $20 \log_{10}((100/5.8)^2) = 49.4 \text{ dB}$ at f_{sw} , comfortably exceeding the 49 dB requirement. Resulting source-side ripple: $16.7 \text{ A}/10^{49.4/20} \approx 56.6 \text{ mA}$ pp.

The choke has a window fill ratio of $\text{fill} = 0.18$ and copper loss of $P_{C_{u,L_2}} = 0.07 \text{ W}$, with little thermal impact.

Table 4: Predicted loss breakdown at $P_{\text{out}} = 160 \text{ W}$, $V_{\text{in}} = 40 \text{ V}$.

Component	Mechanism	Formula	Loss (W)	% of P_{in}
Q_1 (HS)	Conduction	$I_{\text{rms,HS}}^2 R k_T$	0.960	0.59%
Q_1 (HS)	Switching	$\frac{1}{2} V I (t_r + t_f) f$	0.933	0.57%
Q_1 (HS)	Gate drive	$Q_g V_{\text{GS}} f$	0.030	0.02%
Q_1 (HS)	C_{oss}	$\frac{1}{2} C_{\text{oss}} V^2 f$	0.012	0.01%
Q_1 (HS)	Q_{RR} from Q_2	$Q_{\text{RR}} V f$	0.240	0.15%
Q_2 (LS)	Conduction	$I_{\text{rms,LS}}^2 R k_T$	0.448	0.27%
Q_2 (LS)	Gate drive	$Q_g V_{\text{GS}} f$	0.052	0.03%
Q_2 (LS)	Body diode dt	$V_F I (2t_{\text{dt}}) f$	0.064	0.04%
L_1	Cu winding (DC)	$I_{\text{rms,L}}^2 R_{\text{dc}}$	0.570	0.35%
L_1	Core (Steinmetz)	$C_M f^\alpha B_{\text{ac}}^\beta V_e$	0.173	0.11%
C_1	ESR	$I_{\text{rms,out}}^2 \text{ESR}_{\text{eff}}$	0.001	0.00%
C_2	ESR	$I_{\text{rms,in}}^2 \text{ESR}_{\text{eff}}$	0.031	0.02%
L_2	Cu winding (DC)	$I_{\text{in,DC}}^2 R_{\text{dc}}$	0.070	0.04%
Total losses			3.58	2.19%
Predicted efficiency η				97.81%

7 Loss Analysis and Efficiency

Table 4 aggregates the loss contribution of each component at the full-load worst-case. The total system loss is 3.58 W, predicting $\eta = 160/(160 + 3.58) = 97.81\%$.

The dominant loss is HS conduction + switching ($\approx 1.9 \text{ W}$ combined), followed by inductor copper at 0.57 W. The 7.81 pp margin over the 90 % efficiency spec demonstrates that the design choices, synchronous rectification, fast HS device, low- $R_{\text{DS(on)}}$ LS device, polymer caps, gapped ferrite core with adequate copper PCB Heat sinking, come together to provide a low-cost, high efficiency buck converter.

8 LTSpice Simulation and Transfer Function

To verify the closed-loop transient behavior of the converter, I implemented an averaged LTSpice model of the synchronous buck converter. Instead of modeling the full switching waveform, the switching stage was represented using behavioral dependent sources.

The simulated converter used the optimized component values from the design:

$$L_{\text{in}} = 16.2 \mu\text{H}, \quad C_{\text{in}} = 46 \mu\text{F}, \quad (26)$$

$$L_{\text{out}} = 14.4 \mu\text{H}, \quad C_{\text{out}} = 100 \mu\text{F}, \quad (27)$$

$$R_{\text{out}} = 3.6 \Omega. \quad (28)$$

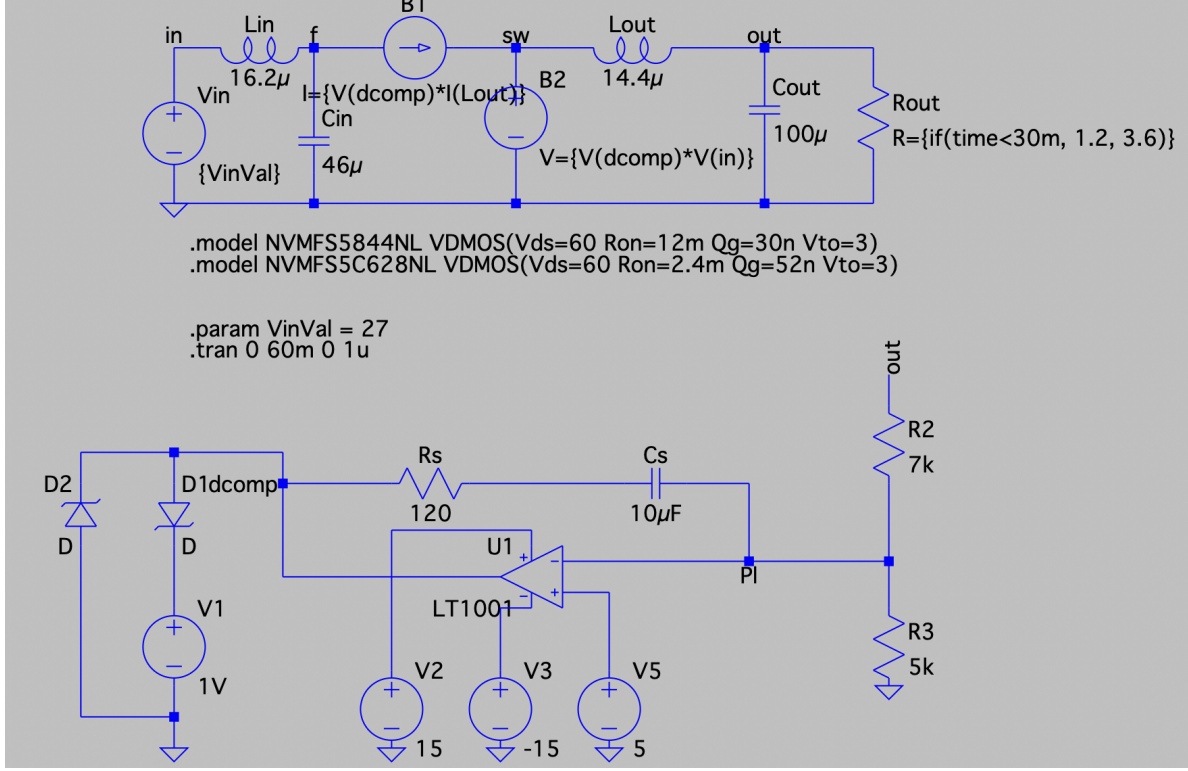


Figure 1: LTSpice averaged model of the synchronous buck converter with the input filter, output filter, and PI compensator.

A PI compensator was implemented with an LT1001 op amp. The output voltage was scaled from 12 V to 5 V using a resistor divider with $R_2 = 7\text{ k}\Omega$ and $R_3 = 5\text{ k}\Omega$. The compensator used $R_s = 120\text{ }\Omega$ and $C_s = 10\text{ }\mu\text{F}$, giving the PI compensator form

$$G_c(s) = P \left(1 + \frac{\omega_L}{s} \right), \quad (29)$$

where

$$P = \frac{R_s}{R_{in}}, \quad \omega_L = \frac{1}{R_s C_s}. \quad (30)$$

The LTSpice transient simulation was run for 60 ms at the worst-case input voltage of 27 V. The output initially overshoots during startup as the compensator and LC filter settle, but the closed-loop system eventually converges to the desired 12 V output. The LTSpice simulation was design with a variable resistor that switched from a 1.2 Ω load to 3.6 Ω after 30ms, representing a change of 120 W, the transient response is shown below, keeping in range of 2 V.

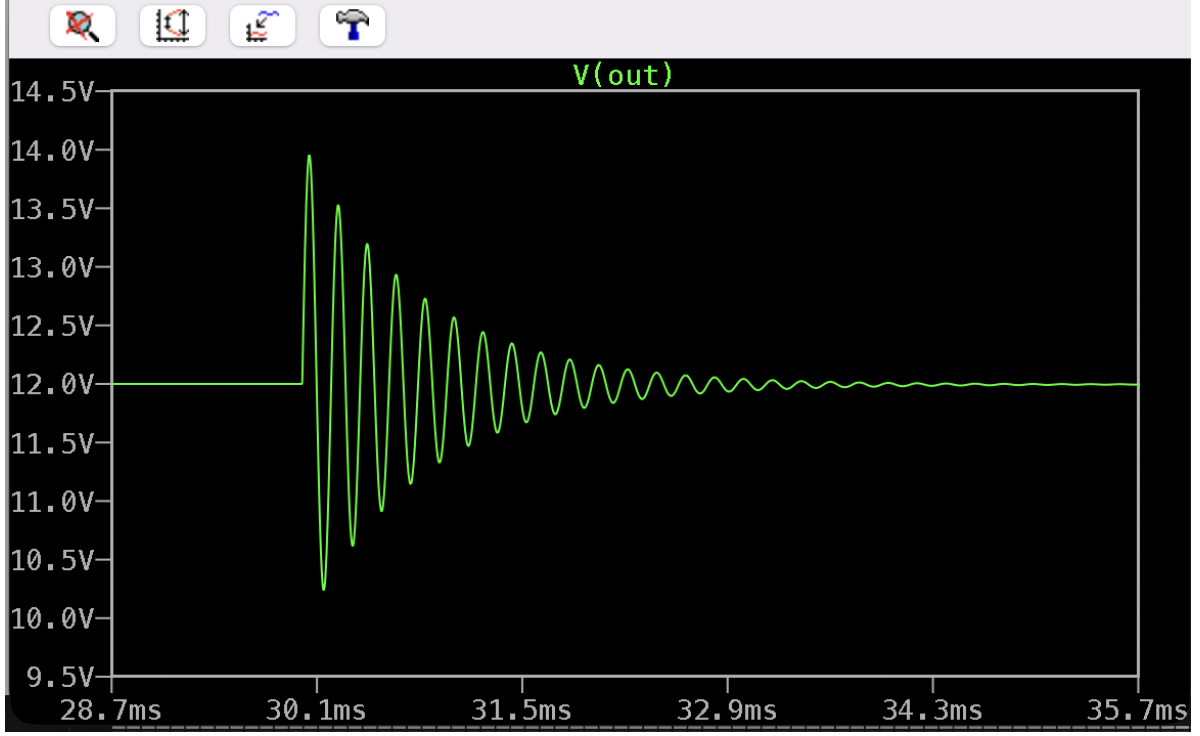


Figure 2: Simulated transient response of the averaged converter model at $V_{in} = 40\text{ V}$.

All four cases of transient reposes were tested for the worst transient response was chosen for figure 2.

Case	V_{in}	R_{before}	R_{after}
0	27 V	$3.6\ \Omega$	$0.9\ \Omega$
1	40 V	$3.6\ \Omega$	$0.9\ \Omega$
2	27 V	$0.9\ \Omega$	$3.6\ \Omega$
3	40 V	$0.9\ \Omega$	$3.6\ \Omega$

9 Bill Of Materials

Table 5 contains the bill of materials for the buck converter. The prices are dominated by the low-ESR output capacitors. Since these capacitors are obsolete, their prices are extremely high. Comparable modern capacitors would reduce the output capacitor cost to around \$40. The cheapest possible converter would cost around \$17, based on the Python optimization output.

Table 5: Bill of materials for the optimal design.

Item	Part	Qty	Unit \$	Ext \$
Q1 HS MOSFET	NVMFS5844NL	1	0.74	0.74
Q2 LS MOSFET	NVMFS5C628NL	1	1.95	1.95
L1 core	RM12/I-3F3-A400	1	3.80	3.80
C1 output cap	106K100CS4	10	49.30	493.00
L2 core	RM10/I-3F3	1	2.55	2.55
C2 input cap	KTS500B335M32N0T00	14	0.80	11.23
			Total	513.27

References

- [1] ROHM Co., Ltd., *Calculation of Power Loss (Synchronous)*, Switching Regulator IC Series Application Note, Rev. 003, Oct. 2016. [Online]. Available: https://fscdn.rohm.com/en/products/databook/applinote/ic/power/switching_regulator/power_loss_appli-e.pdf

Python Optimization Output

=====

OPTIMAL DESIGN (objective = minimise loss)

=====

f_sw : 100 kHz
efficiency (WC) : 97.81 %
total P_loss : 3.58 W
BOM cost (est) : \$513.27

-- MOSFETs (sync buck) --
Q1 (HS): NVMFS5844NL P=2.18 W T_J=83 °C
Q2 (LS): NVMFS5C628NL T_J=58 °C

-- Output inductor L1 --
Core: RM12/I-3F3-A400 N = 6 L = 14.4 µH
dI_pp = 5.83 A, I_rms = 13.44 A, I_pk = 16.25 A
B_pk = 0.267 T (limit 0.30) fill = 0.72
P_Cu = 0.57 W, P_core = 0.173 W, ΔT_core = 17.0 °C

-- Output capacitor C1 --
106K100CS4 (CS), 10× parallel -> C=100 µF, ESR=0.3 mΩ
dV_out = 74.7 mV (limit 75) P_loss = 0.001 W

-- Input inductor L2 --
Core: RM10/I-3F3 N = 2 L = 16.2 µH
fill = 0.18 P_Cu = 0.07 W

-- Input capacitor C2 --
KTS500B335M32N0T00 (NTS), 14× parallel -> C=46 µF
P_loss = 0.031 W
f_r = 5.8 kHz I_source_pp = 56.6 mA (limit 60)

=====

BOM (optimal design)

=====

Item	Part	Qty	Unit \$	Ext \$
Q1 HS MOSFET	NVMFS5844NL	1	0.74	0.74
Q2 LS MOSFET	NVMFS5C628NL	1	1.95	1.95
L1 core	RM12/I-3F3-A400	1	3.80	3.80
C1 output cap	106K100CS4	10	49.30	493.00
L2 core	RM10/I-3F3	1	2.55	2.55
C2 input cap	KTS500B335M32N0T00	14	0.80	11.23

Total 513.27

Top 10 designs by efficiency

f_sw	η %	P_loss	\$ BOM	Q1	Q2	L1 core	L1 uH	C1 uF	nC1	L2 core	L2 uH	C2 uF	nC2			
100k	97.81%	3.58W	\$513.27	NVMFS5844NL	NVMFS5C628NL	RM12/I-3F3-A400	14.40	100	10	RM10/I-3F3	16.20	46	14			
110k	97.74%	3.70W	\$478.37	NVMFS5844NL	NVMFS5C628NL	RM12/I-3F3-A400	14.40	85	18	RM10/I-3F3	16.20	46	14			
120k	97.68%	3.81W	\$423.32	NVMFS5844NL	NVMFS5C628NL	RM12/I-3F3-A400	14.40	72	18	RM8/I-3F3	12.00	46	14			
130k	97.62%	3.91W	\$501.87	NVMFS5844NL	NVMFS5C628NL	RM12/I-3F3-A400	10.00	89	19	RM7/I-3F3	10.00	46	14			
140k	97.54%	4.03W	\$442.02	NVMFS5844NL	NVMFS5C628NL	RM12/I-3F3-A400	10.00	76	19	RM7/I-3F3	10.00	46	14			
150k	97.47%	4.15W	\$397.42	NVMFS5844NL	NVMFS5C628NL	RM12/I-3F3-A400	10.00	68	17	RM7/I-3F3	10.00	46	14			
160k	97.40%	4.27W	\$347.00	NVMFS5844NL	NVMFS5C628NL	RM12/I-3F3-A400	10.00	61	13	RM7/I-3F3	10.00	31	14			
170k	97.33%	4.40W	\$311.95	NVMFS5844NL	NVMFS5C628NL	RM12/I-3F3-A400	10.00	60	6	RM7/I-3F3	10.00	31	14			
180k	97.26%	4.52W	\$289.22	NVMFS5844NL	NVMFS5C628NL	RM12/I-3F3-A400	10.00	48	12	RM12/I-3F3	5.05	46	14			
190k	97.18%	4.64W	\$287.62	NVMFS5844NL	NVMFS5C628NL	RM12/I-3F3-A400	10.00	40	20	RM12/I-3F3	5.05	46	14			

Top 10 designs by BOM cost

f_sw	\$ BOM	η %	P_loss	Q1	Q2	L1 core	L1 uH	C1 uF	nC1	L2 core	L2 uH	C2 uF	nC2			
470k	\$ 17.70	95.26%	7.96W	NVMFS5844NL	NVMFS5C670NL	RM10/I-3F3-A400	6.40	10	15	RM7/I-3F3	2.50	21	14			
480k	\$ 17.70	95.20%	8.07W	NVMFS5844NL	NVMFS5C670NL	RM10/I-3F3-A400	6.40	10	15	RM7/I-3F3	2.50	21	14			
500k	\$ 17.93	95.05%	8.32W	NVMFS5844NL	NVMFS5C670NL	RM10/I-3F3-A400	6.40	9	19	RM7/I-3F3	2.50	21	14			
460k	\$ 18.04	95.32%	7.85W	NVMFS5844NL	NVMFS5C670NL	RM10/I-3F3-A400	6.40	11	16	RM7/I-3F3	2.50	21	14			
330k	\$ 18.09	96.20%	6.32W	NVMFS5844NL	NVMFS5C628NL	RM10/I-3F3-A400	6.40	21	14	RM7/I-3F3	2.50	31	14			
490k	\$ 18.22	95.13%	8.19W	NVMFS5844NL	NVMFS5C670NL	RM10/I-3F3-A400	6.40	9	20	RM7/I-3F3	2.50	21	14			
320k	\$ 18.32	96.27%	6.19W	NVMFS5844NL	NVMFS5C628NL	RM10/I-3F3-A400	6.40	22	15	RM7/I-3F3	2.50	31	14			
450k	\$ 18.39	95.39%	7.74W	NVMFS5844NL	NVMFS5C670NL	RM10/I-3F3-A400	6.40	12	17	RM7/I-3F3	2.50	21	14			
310k	\$ 18.55	96.35%	6.06W	NVMFS5844NL	NVMFS5C628NL	RM10/I-3F3-A400	6.40	24	16	RM7/I-3F3	2.50	31	14			
430k	\$ 18.74	95.52%	7.51W	NVMFS5844NL	NVMFS5C670NL	RM10/I-3F3-A400	6.40	12	18	RM7/I-3F3	2.50	21	14			